

Assignment 3 / Midterm Project - Implement Pipeline for APIs using Jenkins Docker EC2 GitHub and WebHooks

20 points for project and 5 points for Task number, clear picture and comment (if required)

Problem 1

You are part of a DevOps team at a startup called “FoodExpress”, a company that provides an online food ordering platform.

The backend team continuously develops REST APIs that power the mobile and web applications. Currently, developers manually deploy APIs to the production server, which causes several issues:

Slow deployment process

Frequent human errors

Lack of version control in deployments

Difficulty rolling back broken builds

To solve this, the company has decided to implement an automated CI/CD pipeline using modern DevOps tools.

Your DevOps team has been assigned to design and implement a pipeline that automatically builds, tests, containerizes, and deploys the API whenever developers push code to GitHub.

Developer Push Code → Github → Webhook Trigger → Jenkins → Build and Test API → Build Docker Image → Push Image / Deploy → AWS EC2 Server → Running API Container.

Task 1: Create a Public GitHub repository. Clone it to your local computer (Laptop)

```
(base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git remote -v
origin  git@github.com:chanhengmenh/devops-assignment-3.git (fetch)
origin  git@github.com:chanhengmenh/devops-assignment-3.git (push)
(base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3>
```

Task 2: Create a Project folder (install necessary Library) inside Cloned Repository. Write code inside Project folder only for Get APIs in the index.js File. - Screenshot for index.js [Test in the localhost that APIs are working] - Screenshot

```

1  from fastapi import FastAPI, HTTPException
2  from pydantic import BaseModel
3  from typing import Optional
4
5  app = FastAPI(title="FoodExpress API", version="1.0.0")
6
7  # ----- In-memory "database" -----
8  orders = [
9      {"id": 1, "customer": "Alice", "item": "Burger", "quantity": 2, "status": "pending"},
10     {"id": 2, "customer": "Bob", "item": "Pizza", "quantity": 1, "status": "delivered"},
11     {"id": 3, "customer": "Charlie", "item": "Sushi", "quantity": 3, "status": "preparing"},
12 ]
13 next_id = 4
14
15
16 # ----- Request body schema -----
17 class OrderCreate(BaseModel):
18     customer: str
19     item: str
20     quantity: int
21     status: Optional[str] = "pending"
22
23
24 class OrderUpdate(BaseModel):
25     customer: Optional[str] = None
26     item: Optional[str] = None
27     quantity: Optional[int] = None
28     status: Optional[str] = None
29

```

```

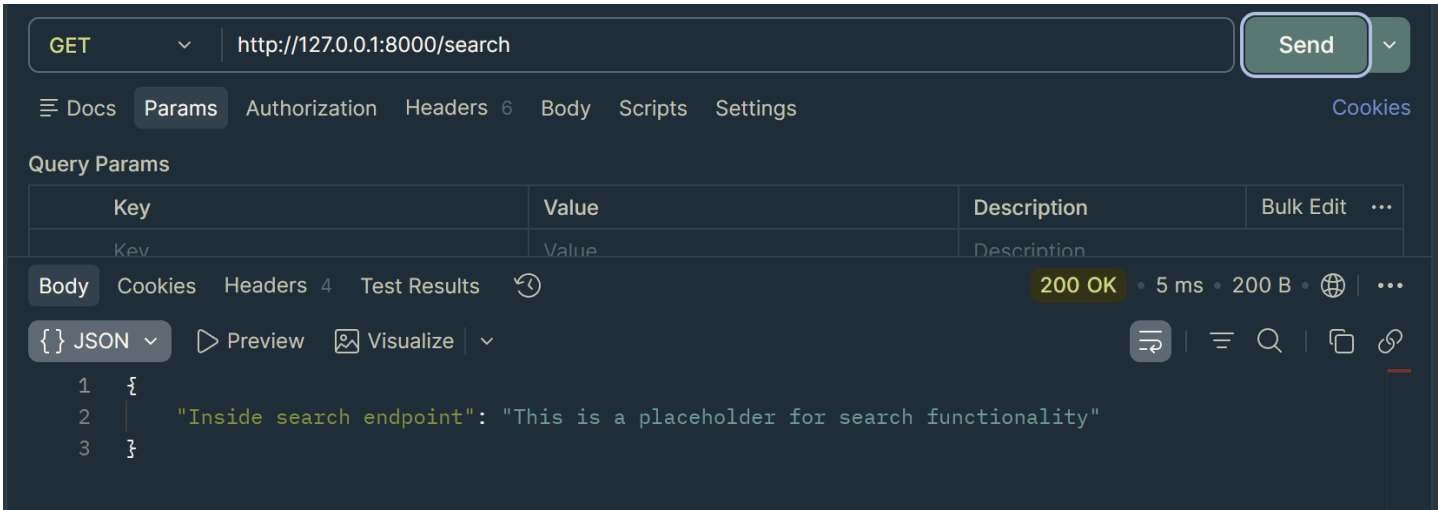
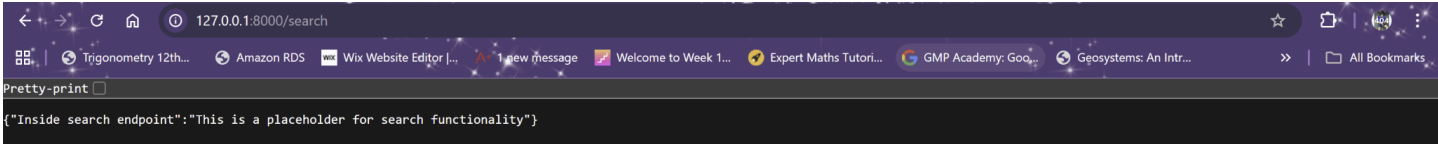
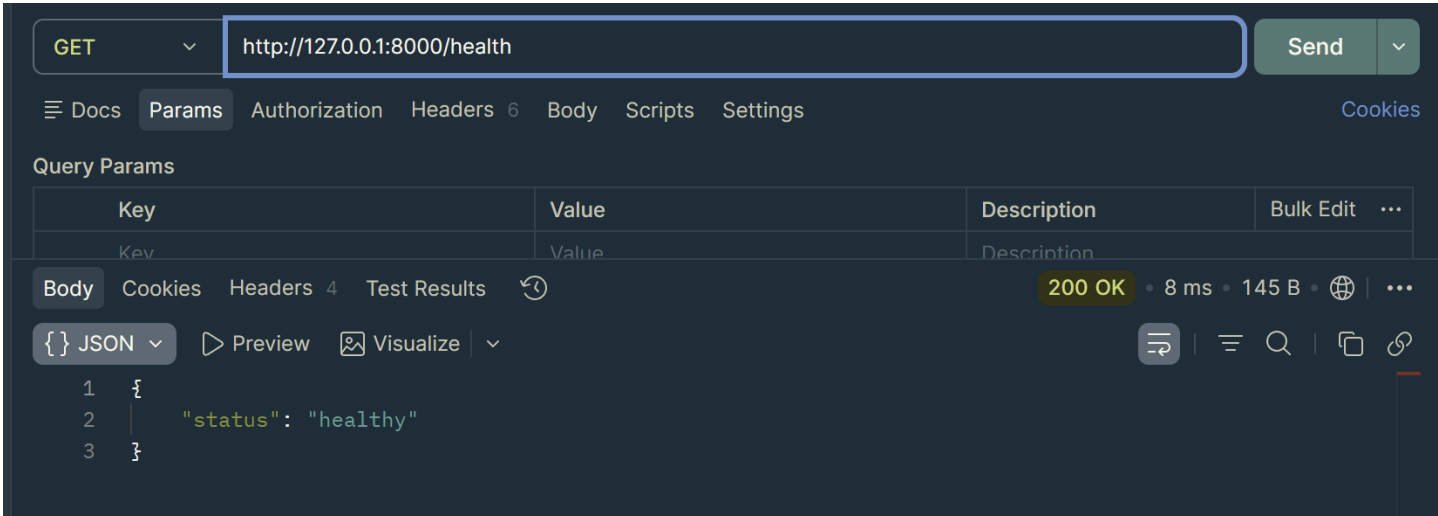
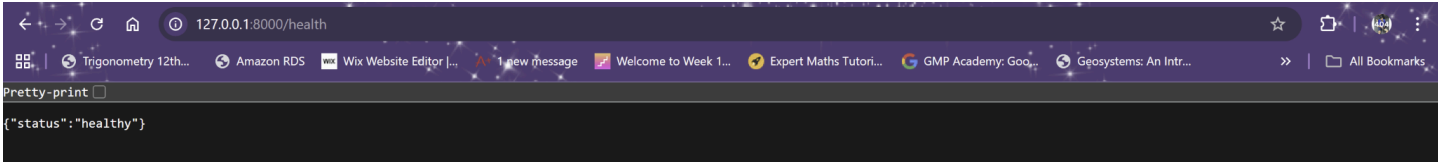
33 @app.get("/")
34 def root():
35     return {"message": "Welcome to FoodExpress API", "version": "1.0.0"}
36
37 Tabnine | Edit | Test | Explain | Document
38 @app.get("/health")
39 def health_check():
40     return {"status": "healthy"}
41
42 Tabnine | Edit | Test | Explain | Document
43 @app.get("/search")
44 def get_orders_count():
45     return {"Inside search endpoint": "This is a placeholder for search functionality"}
46
47 # GET all orders
48 Tabnine | Edit | Test | Explain | Document
49 @app.get("/orders")
50 def get_orders():
51     return {"orders": orders, "total": len(orders)}

```

```

47 # GET all orders
48 Tabnine | Edit | Test | Explain | Document
49 @app.get("/orders")
50 def get_orders():
51     return {"orders": orders, "total": len(orders)}
52
53 # GET single order by ID
54 Tabnine | Edit | Test | Explain | Document
55 @app.get("/orders/{order_id}")
56 def get_order(order_id: int):
57     order = next((o for o in orders if o["id"] == order_id), None)
58     if not order:
59         raise HTTPException(status_code=404, detail=f"Order {order_id} not found")
60     return order
61
62 # POST create new order
63 Tabnine | Edit | Test | Explain | Document
64 @app.post("/orders", status_code=201)
65 def create_order(order: OrderCreate):
66     global next_id
67     new_order = {"id": next_id, **order.dict()}
68     orders.append(new_order)
69     next_id += 1
70     return {"message": "Order created successfully", "order": new_order}

```



127.0.0.1:8000/orders

Trigonometry 12th... Amazon RDS Wix Website Editor [...] new message Welcome to Week 1... Expert Maths Tutori... GMP Academy: Goo... Geosystems: An Intr... All Bookmarks

Pretty-print

```
{
  "orders": [
    {
      "id": 1,
      "customer": "Alice",
      "item": "Burger",
      "quantity": 2,
      "status": "pending"
    },
    {
      "id": 2,
      "customer": "Bob",
      "item": "Pizza",
      "quantity": 1,
      "status": "delivered"
    },
    {
      "id": 3,
      "customer": "Charlie",
      "item": "Sushi",
      "quantity": 3,
      "status": "preparing"
    }
  ],
  "total": 3
}
```

GET

http://127.0.0.1:8000/orders

Send

Docs

Params

Authorization

Headers 6

Body

Scripts

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

Cookies

Headers 4

Test Results

200 OK • 7 ms • 378 B • | ...

JSON

Preview

Visualize

```
1 {
2   "orders": [
3     {
4       "id": 1,
5       "customer": "Alice",
6       "item": "Burger",
7       "quantity": 2,
8       "status": "pending"
9     },
10    {
11      "id": 2,
12      "customer": "Bob",
13      "item": "Pizza",
14      "quantity": 1,
15      "status": "delivered"
16    },
17    {
18      "id": 3,
19      "customer": "Charlie",
20      "item": "Sushi",
21      "quantity": 3,
22      "status": "preparing"
23    }
24  ],
25  "total": 3
26 }
```

127.0.0.1:8000/orders/1

Trigonometry 12th... Amazon RDS Wix Website Editor [...] new message Welcome to Week 1... Expert Maths Tutori... GMP Academy: Goo... Geosystems: An Intr... All Bookmarks

Pretty-print

```
{
  "id": 1,
  "customer": "Alice",
  "item": "Burger",
  "quantity": 2,
  "status": "pending"
}
```

GET

http://127.0.0.1:8000/orders/1

Send

Docs

Params

Authorization

Headers 6

Body

Scripts

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

Cookies

Headers 4

Test Results

200 OK • 10 ms • 200 B •  • 

{}

JSON

Preview

Visualize

1

{

2

" id": 1,

3

"customer": "Alice",

4

"item": "Burger",

5

"quantity": 2,

6

"status": "pending"

7

}

POST

http://127.0.0.1:8000/orders

Send

Docs Params Authorization Headers 8 Body Scripts Settings

Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

Schema Beautify

```
1 {  
2   "id": 4,  
3   "customer": "Chanheng",  
4   "item": "Burger",  
5   "quantity": 2,  
6   "status": "pending"  
7 }
```

Body Cookies Headers 4 Test Results

201 Created • 14 ms • 258 B •

{ } JSON Preview Visualize

| | | |

```
1 {  
2   "message": "Order created successfully",  
3   "order": {  
4     "id": 4,  
5     "customer": "Chanheng",  
6     "item": "Burger",  
7     "quantity": 2,  
8     "status": "pending"  
9   }  
10 }
```

PUT ▼ http://127.0.0.1:8000/orders/1 Send ▼

Docs Params Authorization Headers 8 **Body** ● Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼ Schema Beautify

```
1 {
2   "id": 1,
3   "customer": "Alice",
4   "item": "Burger",
5   "quantity": 2,
6   "status": "Done!"
7 }
```

Body Cookies Headers 4 Test Results ↺ **200 OK** • 5 ms • 248 B • 🌐 | ⋮

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ ⌵ | ≡ | 🔍 | 📋 | 🔗

```
1 {
2   "message": "Order updated successfully",
3   "order": {
4     "id": 1,
5     "customer": "Alice",
6     "item": "Burger",
7     "quantity": 2,
8     "status": "Done!"
9   }
10 }
```

DELETE ▼ http://127.0.0.1:8000/orders/1 Send ▼

Docs Params Authorization **Headers** 8 **Body** ● Scripts Settings Cookies

Body Cookies Headers 4 Test Results ↺ **200 OK** • 5 ms • 167 B • 🌐 | ⋮

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ ⌵ | ≡ | 🔍 | 📋 | 🔗

```
1 {
2   "message": "Order 1 deleted successfully"
3 }
```

Task 3: Add and commit it. Push it in the GitHub. - Screenshot

```

• (base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git add .
warning: in the working copy of 'FoodExpressAPI/main.py', LF will be replaced by CRLF the next time Git touches it
• (base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git commit -m 'first commit'
[main 4b60aee] first commit
 2 files changed, 23 insertions(+), 19 deletions(-)
• (base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.77 KiB | 904.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com-personal:chanhengmenh/devops-assignment-3.git
 893c4f4..4b60aee  main -> main
• (base) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3>

```

The screenshot shows a GitHub repository page for 'devops-assignment-3' (Public). The repository has 1 branch (main) and 0 tags. The commit history shows a 'first commit' by 'chanhengmenh' with 5 commits. The file list includes 'FoodExpressAPI', '.gitignore', 'Jenkinsfile', and 'README.md'. The README file is open, showing the title 'FoodExpress API — DevOps Assignment 3' and a description: 'A FastAPI-based food ordering REST API with a fully automated CI/CD pipeline using Jenkins and Docker, deployed on AWS EC2.' The table of contents includes links to 'Project Structure', 'Tech Stack', 'API Endpoints', and 'Running Locally'. The right sidebar shows 'About' (No description), 'Releases' (No releases published), 'Packages' (No packages published), 'Contributors' (1 contributor: 'chanhengmenh'), and 'Languages' (Python 92.7%, Dockerfile 7.3%).

Task 4: Create EC2 instance using Ubuntu and create your own keypair. - Screenshot Public IP only

The screenshot shows the AWS Management Console details for an EC2 instance named 'i-086faaa54677af606' (devops-assignment-3). The instance is in the 'Running' state. The public IPv4 address is '54.91.214.38'. The private IPv4 address is '172.31.20.96'. The public DNS is 'ec2-54-91-214-38.compute-1.amazonaws.com'. The instance type is 't2.medium'. The VPC ID is 'vpc-0f3d67439c854476c'. The AWS Compute Optimizer finding is 'Opt-in to AWS Compute Optimizer for recommendations'.

i-086faaa54677af606 (devops-assignment-3)		
Instance ID i-086faaa54677af606	Public IPv4 address 54.91.214.38 open address	Private IPv4 addresses 172.31.20.96
IPv6 address -	Instance state Running	Public DNS ec2-54-91-214-38.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-20-96.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-20-96.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.medium	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations.
Auto-assigned IP address 54.91.214.38 [Public IP]	VPC ID vpc-0f3d67439c854476c	

Task 5: Install Jenkins in the EC2 instance and configure necessary ports. -Some Screenshot

```
ubuntu@ip-172-31-20-96: ~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0f20cab2593fd46d4	HTTP	TCP	80	Custom	0.0.0.0/0	Delete
sgr-041ea6c7563377dd9	SSH	TCP	22	Custom	0.0.0.0/0	Delete
-	Custom TCP	TCP	8080	Any...	0.0.0.0/0	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Task 6.1 - Create a new pipeline and configure it for webhooks. - Screenshots for all settings and each build stage and steps

Jenkins / All / New Item

New Item

Enter an item name

FoodExpress-Pipeline

Select an item type

-  **Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
-  **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
-  **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
-  **Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.
-  **Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

[OK](#)

Advanced

Plain text Preview

☒ Discard old builds ?

Strategy

Log Rotation

Days to keep builds

if not empty, build records are only kept up to this number of days

6

Max # of builds to keep

if not empty, only up to this number of build records are kept

3

Advanced

☐ Discard old builds ?

☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume if the controller restarts

☒ GitHub project

Project url ?

https://github.com/chanhengmenh/devops-assignement-3.git

Advanced

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built ?

☐ Build periodically ?

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Configure

- General
- Triggers
- Pipeline**
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/chanhengmenh/devops-assignment-3.git

Credentials ?

- none -

+ Add

Advanced

+ Add Repository

Branches to build ?

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Script Path ?

Jenkinsfile

☒ Lightweight checkout ?

[Pipeline Syntax](#)

```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Clone') {
6       steps {
7         echo 'Cloning the code from GitHub...'
8         git url: 'https://github.com/chanhengmenh/devops-assignment-3.git', branch: 'main'
9         echo 'Cloning Done'
10      }
11    }
12  }
13
14  post {
15    failure {
16      echo 'Pipeline failed. Check the Console Output for details.'
17    }
18    success {
19      echo 'Deployment Successful! FoodExpress APIs are live.'
20    }
21  }
22 }
23
24
25
```

Task 6.2 - Manually Build it to check if your pipeline is working or not for each stage.

Clone Stage:

Jenkins

FoodExpress-Pipeline

#2

Search

Settings

User

Status

Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

✓ #2 (Mar 17, 2026, 6:00:43 PM)

Add description

Keep this build forever

Started by user [Hello](#)

Started 10 sec ago

Took [5.7 sec](#)

⌚

This run spent:

- 18 ms waiting;
- 5.7 sec build duration;
- 5.7 sec total from scheduled to completion.

git

Revision: 622391cba19a5ab872f9adf69a9995b4a257de06

Repository: <https://github.com/chanhengmenh/devops-assignment-3.git>

- refs/remotes/origin/main

</>

No changes.

Jenkins

FoodExpress-Pipeline

#2

Search

Settings

User

Status

Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Console

Download

Copy

View as plain text

Started by user Hello

Obtained Jenkinsfile from [git https://github.com/chanhengmenh/devops-assignment-3.git](https://github.com/chanhengmenh/devops-assignment-3.git)

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/FoodExpress-Pipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository <https://github.com/chanhengmenh/devops-assignment-3.git>

> git init /var/lib/jenkins/workspace/FoodExpress-Pipeline # timeout=10

Fetching upstream changes from <https://github.com/chanhengmenh/devops-assignment-3.git>

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- <https://github.com/chanhengmenh/devops-assignment-3.git> +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url <https://github.com/chanhengmenh/devops-assignment-3.git> # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 622391cba19a5ab872f9adf69a9995b4a257de06 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 622391cba19a5ab872f9adf69a9995b4a257de06 # timeout=10

Commit message: "fix Jenkinsfile"

First time build. Skipping changelog.

[Pipeline] }

[Pipeline] // stage

[Pipeline] withEnv

Jenkins

FoodExpress-Pipeline

#2

Search

Settings

User

Console

Download

Copy

View as plain text

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Clone)

[Pipeline] echo

Cloning the code from GitHub...

[Pipeline] git

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url <https://github.com/chanhengmenh/devops-assignment-3.git> # timeout=10

Fetching upstream changes from <https://github.com/chanhengmenh/devops-assignment-3.git>

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- <https://github.com/chanhengmenh/devops-assignment-3.git> +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 622391cba19a5ab872f9adf69a9995b4a257de06 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 622391cba19a5ab872f9adf69a9995b4a257de06 # timeout=10

> git branch -a -v --no-abbrev # timeout=10

> git checkout -b main 622391cba19a5ab872f9adf69a9995b4a257de06 # timeout=10

Commit message: "fix Jenkinsfile"

[Pipeline] echo

Cloning Done

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Declarative: Post Actions)

[Pipeline] echo

Deployment Successful! FoodExpress APIs are live.

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

Copy Stage:

```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Clone') {
7       steps {
8         echo 'Cloning the code from GitHub...'
9         git url: 'https://github.com/chanhengmenh/devops-assignment-3.git', branch: 'main'
10        echo 'Cloning Done'
11      }
12    }
13
14    stage('Copy') {
15      steps {
16        echo 'Copying files to /home/ubuntu/current...'
17        sh 'cp -r FoodExpressAPI /home/ubuntu/current'
18        echo 'Copy Done'
19      }
20    }
21  }
22
23  post {
24    failure {
25      echo 'Pipeline failed. Check the Console Output for details.'
26    }
27    success {
28      echo 'Deployment Successful! FoodExpress APIs are live.'
29    }
30  }
31 }
32
33 }
```

 **Jenkins** / FoodExpress-Pipeline ▾ / #3

Status

 **#3 (Mar 17, 2026, 6:08:57 PM)**

 Add description

Keep this build forever

</> Changes

 Console Output

 Edit Build Information

 Delete build '#3'

 Timings

 Git Build Data

 Pipeline Overview

 Restart from Stage

 Replay

 Pipeline Steps

 Started by user Hello

 This run spent:

- 10 ms waiting;
- 2.2 sec build duration;
- 2.2 sec total from scheduled to completion.

 **git**

Revision: d872ab612e246617d8e72043f720ba5dd9c6fb72

Repository: <https://github.com/chanhengmenh/devops-assignment-3.git>

- refs/remotes/origin/main

  fix Jenkinsfile

d872ab6

chanhengmenh at 6:08 PM 3/17/26

Started 6.2 sec ago

Took 2.2 sec

 **Jenkins** / FoodExpress-Pipeline ▾ / #3 ▾

Status

 Changes

Console Output

Edit Build Information

Delete build '#3'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Download

Copy

View as plain text

Started by user Hello

Obtained Jenkinsfile from git <https://github.com/chanhengmenh/devops-assignment-3.git>

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/FoodExpress-Pipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10

Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision d872ab612e246617d8e72043f720ba5dd9c6fb72 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f d872ab612e246617d8e72043f720ba5dd9c6fb72 # timeout=10

Commit message: "fix Jenkinsfile"

> git rev-list --no-walk 622391cba19a5ab872f9adf69a9995b4a257de06 # timeout=10

[Pipeline] }

[Pipeline] // stage

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Clone)

```
Jenkins / FoodExpress-Pipeline / #3

[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone)
[Pipeline] echo
Cloning the code from GitHub...
[Pipeline] git
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10
Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision d872ab612e246617d8e72043f720ba5dd9c6fb72 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f d872ab612e246617d8e72043f720ba5dd9c6fb72 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git branch -D main # timeout=10
> git checkout -b main d872ab612e246617d8e72043f720ba5dd9c6fb72 # timeout=10
Commit message: "fix Jenkinsfile"
[Pipeline] echo
Cloning Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Copy)
[Pipeline] echo
Copying files to /home/ubuntu/current...
[Pipeline] sh
+ cp -r FoodExpressAPI /home/ubuntu/current
[Pipeline] echo
```

```
Jenkins / FoodExpress-Pipeline / #3

> git branch -a -v --no-abbrev # timeout=10
> git branch -D main # timeout=10
> git checkout -b main d872ab612e246617d8e72043f720ba5dd9c6fb72 # timeout=10
Commit message: "fix Jenkinsfile"
[Pipeline] echo
Cloning Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Copy)
[Pipeline] echo
Copying files to /home/ubuntu/current...
[Pipeline] sh
+ cp -r FoodExpressAPI /home/ubuntu/current
[Pipeline] echo
Copy Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful! FoodExpress APIs are live.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

Jenkins 2.541.2
```

Task 6.3 - Install necessary software if required

Jenkins / Manage Jenkins / Plugins

Plugins

Search installed plugins

Name	Health	Enabled
Ant Plugin 520.vd082ecfb_16a_9 Adds Apache Ant support to Jenkins Report an issue with this plugin	96	☒
Apache HttpComponents Client 4.x API Plugin 4.5.14-269.vfa_2321039a_83 Bundles Apache HttpComponents Client 4.x and allows it to be used by Jenkins plugins. Report an issue with this plugin	92	☒
ASM API Plugin 9.9.1-189.vb_5ef2964da_91 This plugin provides the ASM APIs (v9.9.1) for other plugins. Report an issue with this plugin	100	☒
Bootstrap 5 API Plugin 5.3.8-895.v4d0d8e47fea_d Provides Bootstrap 5 for Jenkins Plugins. Bootstrap is (according to their self-perception) the world's most popular front-end component library to build responsive, mobile-first projects on the web. Report an issue with this plugin	96	☒
bouncycastle API Plugin 2.30.1.83-289.v8426fcd19371 This plugin provides a stable API to Bouncy Castle related tasks. Report an issue with this plugin	100	☒
Branch API Plugin 2.1280.v0d4e5b_b_460ef This plugin provides an API for multiple branch based projects. Report an issue with this plugin	100	☒
Build Timeout 1.40 This plugin allows you to automatically terminate a build if it's taking too long. Report an issue with this plugin	100	☒
Caffeine API Plugin 3.2.3-194.v31a_b_f7a_b_5a_81 Caffeine api plugin for use by other Jenkins plugins. Report an issue with this plugin	100	☒

Task 6.4 - If required, set Jenkins Permission - Screenshot

```
ubuntu@ip-172-31-20-96:~$ sudo usermod -aG ubuntu jenkins
ubuntu@ip-172-31-20-96:~$ sudo chmod -R 777 /home/ubuntu
ubuntu@ip-172-31-20-96:~$

ubuntu@ip-172-31-20-96:~$ sudo usermod -aG docker jenkins
ubuntu@ip-172-31-20-96:~$ sudo systemctl restart jenkins
```

Task 7 - Do needful to create a docker image. - Screenshot

→ Install Docker:

ubuntu@ip-172-31-20-96: ~
ubuntu@ip-172-31-20-96:~\$ sudo apt remove \$(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-doc podman-docker containerd run c | cut -f1)

ubuntu@ip-172-31-20-96:~\$ # Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: \$(. /etc/os-release && echo "\${UBUNTU_CODENAME}~\${VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF
sudo apt update

ubuntu@ip-172-31-20-96:~\$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
docker-ce-rootless-extras libsllp0 pigz slirp4netns
Suggested packages:
cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libsllp0 pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 111 not upgraded.
Need to get 97.2 MB of archives.
After this operation, 393 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libsllp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.2.2-1~ubuntu.24.04~noble [23.6 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.3.0-1~ubuntu.24.04~noble [16.4 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.3.0-1~ubuntu.24.04~noble [22.6 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.31.1-1~ubuntu.24.04~noble [20.3 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.3.0-1~ubuntu.24.04~noble [6390 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.1.0-1~ubuntu.24.04~noble [7847 kB]
Fetched 97.2 MB in 1s (78.7 MB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 86626 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.2.2-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking containerd.io (2.2.2-1~ubuntu.24.04~noble) ...

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-96:~\$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-03-17 17:34:49 UTC; 27s ago
TriggeredBy: ● docker.socket
Docs: https://docs.docker.com
Main PID: 5806 (dockerd)
Tasks: 9
Memory: 27.3M (peak: 27.7M)
CPU: 367ms
CGroup: /system.slice/docker.service
└─5806 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.193845055Z" level=info msg="Restoring containers: start."
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.226116116Z" level=info msg="Deleting nftables IPv4 rules" error="exit stat
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.232050967Z" level=info msg="Deleting nftables IPv6 rules" error="exit stat
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.536142375Z" level=info msg="Loading containers: done."
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.546027516Z" level=info msg="Docker daemon" commit=83bca51 containerd-snaps
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.546152031Z" level=info msg="Initializing buildkit"
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.590539013Z" level=info msg="Completed buildkit initialization"
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.600808511Z" level=info msg="Daemon has completed initialization"
Mar 17 17:34:49 ip-172-31-20-96 dockerd[5806]: time="2026-03-17T17:34:49.600876698Z" level=info msg="API listen on /run/docker.sock"
Mar 17 17:34:49 ip-172-31-20-96 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)

File Edit Selection View Go Run ...
devops-assignment-3
MANUAL.md Dockerfile main.py 2 Jenkinsfile
FoodExpressAPI > Dockerfile > ...
1 FROM python:3.11-slim
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 8000
11
12 CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
13

Add 'Build Docker Image' Stage:

```
19 }
20 }
21 }
22 stage('Build') {
23     steps {
24         echo 'Building Docker image...'
25         sh 'docker build -t foodexpress/fastapi:v1.0 /home/ubuntu/current/FoodExpressAPI'
26         echo 'Build Done'
27     }
28 }
29 }
30 }
31 }
```

→ Manually Build:

Jenkins / FoodExpress-Pipeline #5

Status

Changes

Console Output

Edit Build Information

Delete build '#5'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Started by user Hello

This run spent:

- 48 ms waiting;
- 20 sec build duration;
- 20 sec total from scheduled to completion.

Revision: 60aba73cadee870514ae93df277e3b6edc8b7129
Repository: <https://github.com/chanhengmenh/devops-assignment-3.git>

- refs/remotes/origin/main

No changes.

Add description

Keep this build forever

Started 33 sec ago
Took 20 sec

Jenkins / FoodExpress-Pipeline #5

Status

Changes

Console Output

Edit Build Information

Delete build '#5'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Console

Download

Copy

View as plain text

```
Started by user Hello
Obtained Jenkinsfile from git https://github.com/chanhengmenh/devops-assignment-3.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/FoodExpress-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10
Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 60aba73cadee870514ae93df277e3b6edc8b7129 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 60aba73cadee870514ae93df277e3b6edc8b7129 # timeout=10
Commit message: "fix Jenkinsfile"
> git rev-list --no-walk 60aba73cadee870514ae93df277e3b6edc8b7129 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone)
```


Jenkins / FoodExpress-Pipeline #5

[Pipeline] echo
Cloning the code from GitHub...
[Pipeline] git
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10
Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/
origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 60aba73cadee870514ae93df277e3b6edc8b7129 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 60aba73cadee870514ae93df277e3b6edc8b7129 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git branch -D main # timeout=10
> git checkout -b main 60aba73cadee870514ae93df277e3b6edc8b7129 # timeout=10
Commit message: "fix Jenkinsfile"
[Pipeline] echo
Cloning Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Copy)
[Pipeline] echo
Copying files to /home/ubuntu/current...
[Pipeline] sh
+ cp -r FoodExpressAPI /home/ubuntu/current
[Pipeline] echo
Copy Done
[Pipeline] }

Jenkins / FoodExpress-Pipeline #5

[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building Docker image...
[Pipeline] sh
+ docker build -t foodexpress/fastapi:v1.0 /home/ubuntu/current/FoodExpressAPI
#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 242B done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/python:3.11-slim
#2 DONE 0.4s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [internal] load build context
#4 transferring context: 8.75kB done
#4 DONE 0.0s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 resolve docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a 0.0s done
#5 DONE 0.1s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 sha256:7524aba74b8cc45ff3cf2f089fd9a638dfd58b001763194d0819c8d7c75d598f 250B / 250B 0.1s done
#5 sha256:6944007a5b45fe41656cd72b1691b1419bf9d39641e7afbc937f3a3abd07e575 1.29MB / 1.29MB 0.1s done
#5 sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88 11.53MB / 14.37MB 0.3s
#5 sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 6.29MB / 29.78MB 0.2s
#5 sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88 14.37MB / 14.37MB 0.4s done
#5 sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 29.78MB / 29.78MB 0.5s



Jenkins

/ FoodExpress-Pipeline

#5



```
#5 sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88 14.37MB / 14.37MB 0.4s done
#5 sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 29.78MB / 29.78MB 0.5s
#5 sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 29.78MB / 29.78MB 0.5s done
#5 extracting sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f
#5 extracting sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 1.1s done
#5 DONE 1.8s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 extracting sha256:6944007a5b45fe41656cd72b1691b1419bf9d39641e7afbc937f3a3abd07e575 0.1s done
#5 DONE 1.9s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 extracting sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88
#5 extracting sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88 0.7s done
#5 DONE 2.6s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 extracting sha256:7524aba74b8cc45ff3cf2f089fd9a638dfd58b001763194d0819c8d7c75d598f 0.0s done
#5 DONE 2.6s

#6 [2/5] WORKDIR /app
#6 DONE 1.3s

#7 [3/5] COPY requirements.txt .
#7 DONE 0.0s

#8 [4/5] RUN pip install --no-cache-dir -r requirements.txt
#8 2.704 Collecting fastapi (from -r requirements.txt (line 1))
#8 2.729 Downloading fastapi-0.135.1-py3-none-any.whl.metadata (30 kB)
#8 2.780 Collecting uvicorn (from -r requirements.txt (line 2))
#8 2.784 Downloading uvicorn-0.42.0-py3-none-any.whl.metadata (6.7 kB)
#8 2.998 Collecting pydantic (from -r requirements.txt (line 3))
#8 3.002 Downloading pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
#8 3.008 _____ 90.6/90.6 kB 24.1 MB/s eta 0:00:00
#8 3.100 Collecting starlette>=0.46.0 (from fastapi->-r requirements.txt (line 1))
#8 3.104 Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
```



Jenkins

/ FoodExpress-Pipeline

#5



```
#8 3.008 _____ 90.6/90.6 kB 24.1 MB/s eta 0:00:00
#8 3.100 Collecting starlette>=0.46.0 (from fastapi->-r requirements.txt (line 1))
#8 3.104 Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
#8 3.153 Collecting typing-extensions>=4.8.0 (from fastapi->-r requirements.txt (line 1))
#8 3.158 Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
#8 3.176 Collecting typing-inspection>=0.4.2 (from fastapi->-r requirements.txt (line 1))
#8 3.180 Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
#8 3.196 Collecting annotated-doc>=0.0.2 (from fastapi->-r requirements.txt (line 1))
#8 3.199 Downloading annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
#8 3.245 Collecting click>=7.0 (from uvicorn->-r requirements.txt (line 2))
#8 3.250 Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
#8 3.269 Collecting h11>=0.8 (from uvicorn->-r requirements.txt (line 2))
#8 3.273 Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
#8 3.299 Collecting annotated-types>=0.6.0 (from pydantic->-r requirements.txt (line 3))
#8 3.303 Downloading annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
#8 4.430 Collecting pydantic-core==2.41.5 (from pydantic->-r requirements.txt (line 3))
#8 4.434 Downloading pydantic_core-2.41.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)
#8 4.499 Collecting anyio<5,>=3.6.2 (from starlette>=0.46.0->fastapi->-r requirements.txt (line 1))
#8 4.502 Downloading anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
#8 4.549 Collecting idna>=2.8 (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi->-r requirements.txt (line 1))
#8 4.554 Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
#8 4.573 Downloading fastapi-0.135.1-py3-none-any.whl (116 kB)
#8 4.576 _____ 117.0/117.0 kB 155.3 MB/s eta 0:00:00
#8 4.580 Downloading uvicorn-0.42.0-py3-none-any.whl (68 kB)
#8 4.583 _____ 68.8/68.8 kB 145.9 MB/s eta 0:00:00
#8 4.588 Downloading pydantic-2.12.5-py3-none-any.whl (463 kB)
#8 4.593 _____ 463.6/463.6 kB 172.9 MB/s eta 0:00:00
#8 4.597 Downloading pydantic_core-2.41.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.1 MB)
#8 4.612 _____ 2.1/2.1 MB 165.2 MB/s eta 0:00:00
#8 4.616 Downloading annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
#8 4.621 Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
#8 4.626 Downloading click-8.3.1-py3-none-any.whl (108 kB)
#8 4.629 _____ 108.3/108.3 kB 180.3 MB/s eta 0:00:00
#8 4.633 Downloading h11-0.16.0-py3-none-any.whl (37 kB)
#8 4.638 Downloading starlette-0.52.1-py3-none-any.whl (74 kB)
```

 Jenkins

FoodExpress-Pipeline

#5

Q

⚙️

👤

```
#8 4.647 _____ 44.6/44.6 kB 152.9 MB/s eta 0:00:00
#8 4.651 Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
#8 4.656 Downloading anyio-4.12.1-py3-none-any.whl (113 kB)
#8 4.659 _____ 113.6/113.6 kB 163.1 MB/s eta 0:00:00
#8 4.665 Downloading idna-3.11-py3-none-any.whl (71 kB)
#8 4.668 _____ 71.0/71.0 kB 173.9 MB/s eta 0:00:00
#8 4.877 Installing collected packages: typing-extensions, idna, h11, click, annotated-types, annotated-doc, uvicorn, typing-
inspection, pydantic-core, anyio, starlette, pydantic, fastapi
#8 6.074 Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 fastapi-0.135.1 h11-0.16.0
idna-3.11 pydantic-2.12.5 pydantic-core-2.41.5 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.42.0
#8 6.074 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system
package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
#8 6.168
#8 6.168 [notice] A new release of pip is available: 24.0 -> 26.0.1
#8 6.168 [notice] To update, run: pip install --upgrade pip
#8 DONE 6.5s

#9 [5/5] COPY . .
#9 DONE 0.1s

#10 exporting to image
#10 exporting layers
#10 exporting layers 1.7s done
#10 exporting manifest sha256:c402c447069ebb198b5d37fc19b942f2c48a122f793eae74058a50c67587ad03
#10 exporting manifest sha256:c402c447069ebb198b5d37fc19b942f2c48a122f793eae74058a50c67587ad03 0.0s done
#10 exporting config sha256:dcae8e48d7bb9da67fae157fdbd36f0dd2f986f953b3c9ddc56aa9e9387b9217 0.0s done
#10 exporting attestation manifest sha256:e874aa1bcaf679d7c1ec818c08e9323bbf61befa08033df8c93716dc8cf1185c 0.0s done
#10 exporting manifest list sha256:e9c5af558f95c61bbec975efd201da3ca588e2187b74b1ef61ed82f40c288ab3 0.0s done
#10 naming to docker.io/foodexpress/fastapi:v1.0 done
#10 unpacking to docker.io/foodexpress/fastapi:v1.0
#10 unpacking to docker.io/foodexpress/fastapi:v1.0 0.6s done
#10 DONE 2.4s
[Pipeline] echo
Build Done
[Pipeline] }
```

Clear

Environment

Docker Docs

Jenkins

Deployment - 3 - 0

2 | us-east-1

Connect | us

```
#10 exporting manifest sha256:c402c447069ebb198b5d37fc19b942f2c48a122f793eae74058a50c67587ad03 0.0s done
#10 exporting config sha256:dcae8e48d7bb9da67fae157fdbd36f0dd2f986f953b3c9ddc56aa9e9387b9217 0.0s done
#10 exporting attestation manifest sha256:e874aa1bcaf679d7c1ec818c08e9323bbf61befa08033df8c93716dc8cf1185c 0.0s done
#10 exporting manifest list sha256:e9c5af558f95c61bbec975efd201da3ca588e2187b74b1ef61ed82f40c288ab3 0.0s done
#10 naming to docker.io/foodexpress/fastapi:v1.0 done
#10 unpacking to docker.io/foodexpress/fastapi:v1.0
#10 unpacking to docker.io/foodexpress/fastapi:v1.0 0.6s done
#10 DONE 2.4s
[Pipeline] echo
Build Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful! FoodExpress APIs are live.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.2

ubuntu@ip-172-31-20-96:~\$ sudo docker images

Info →

In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
foodexpress/fastapi:v1.0	e9c5af558f95	225MB	54.6MB	

ubuntu@ip-172-31-20-96:~\$

Task 8 - Do needful to Deploy it inside the EC2. - Screenshot

EC2 > Security Groups > sg-0abf0a6c6b588b5d9 - launch-wizard-1 > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional		
sgr-04da62343c8e72380	Custom TCP	TCP	8080	Custom	Q	0.0.0.0/0	Delete
sgr-0f20cab2593fd46d4	HTTP	TCP	80	Custom	Q	0.0.0.0/0	Delete
sgr-041ea6c7563377dd9	SSH	TCP	22	Custom	Q	0.0.0.0/0	Delete
-	Custom TCP	TCP	8000	Any...	Q	0.0.0.0/0	Delete

Add rule

Rules with source of 0.0.0.0/0 or ::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

```
26     echo 'Build Done'
27   }
28 }
29
30 stage('Run Image As Container') {
31   steps {
32     echo 'Deploying container...'
33     sh '''
34       docker stop foodexpress-container || true
35       docker rm foodexpress-container || true
36       docker run --name foodexpress-container -d -p 8000:8000 foodexpress/Fastapi:v1.0
37     '''
38     echo 'FoodExpress FastAPI is running inside Docker on port 8000'
39   }
40 }
41
42
43
```

Jenkins / FoodExpress-Pipeline #8

Status

Changes

Console Output

Edit Build Information

Delete build '#8'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

#8 (Mar 17, 2026, 7:23:12 PM)

Started by user Hello

This run spent:

- 11 ms waiting;
- 3.6 sec build duration;
- 3.6 sec total from scheduled to completion.

git

Revision: de82d3735d0ce1da819c0da3673de5e63f8971b2

Repository: https://github.com/chanhengmenh/devops-assignment-3.git

- refs/remotes/origin/main

No changes.

Add description

Keep this build forever

Started 1 min 44 sec ago

Took 3.6 sec

Jenkins

FoodExpress-Pipeline

#8

Search

Settings

User

Status

Changes

Console Output

Edit Build Information

Delete build '#8'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

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Pipeline Steps

Workspaces

Previous Build

Console

Download

Copy

View as plain text

Started by user Hello

Obtained Jenkinsfile from git https://github.com/chanhengmenh/devops-assignment-3.git

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in /var/lib/jenkins/workspace/FoodExpress-Pipeline

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Declarative: Checkout SCM)

[Pipeline] checkout

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10

Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision de82d3735d0ce1da819c0da3673de5e63f8971b2 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f de82d3735d0ce1da819c0da3673de5e63f8971b2 # timeout=10

Commit message: "fix Jenkinsfile"

> git rev-list --no-walk de82d3735d0ce1da819c0da3673de5e63f8971b2 # timeout=10

[Pipeline] }

[Pipeline] // stage

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Clone)

Jenkins

FoodExpress-Pipeline

#8

Search

Settings

User

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Changes

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Edit Build Information

Delete build '#8'

Timings

Git Build Data

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Previous Build

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Download

Copy

View as plain text

[Pipeline] echo

Cloning the code from GitHub...

[Pipeline] git

Selected Git installation does not exist. Using Default

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/FoodExpress-Pipeline/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/chanhengmenh/devops-assignment-3.git # timeout=10

Fetching upstream changes from https://github.com/chanhengmenh/devops-assignment-3.git

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git fetch --tags --force --progress -- https://github.com/chanhengmenh/devops-assignment-3.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision de82d3735d0ce1da819c0da3673de5e63f8971b2 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f de82d3735d0ce1da819c0da3673de5e63f8971b2 # timeout=10

> git branch -a -v --no-abbrev # timeout=10

> git branch -D main # timeout=10

> git checkout -b main de82d3735d0ce1da819c0da3673de5e63f8971b2 # timeout=10

Commit message: "fix Jenkinsfile"

[Pipeline] echo

Cloning Done

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Copy)

[Pipeline] echo

Copying files to /home/ubuntu/current...

[Pipeline] sh

+ cp -r FoodExpressAPI /home/ubuntu/current

[Pipeline] echo

Copy Done

[Pipeline] }

Jenkins / FoodExpress-Pipeline #8

```
[Pipeline] /
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building Docker image...
[Pipeline] sh
+ docker build -t foodexpress/fastapi:v1.0 /home/ubuntu/current/FoodExpressAPI
#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 242B done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/python:3.11-slim
#2 DONE 0.1s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [internal] load build context
#4 transferring context: 8.75kB done
#4 DONE 0.0s

#5 [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a
#5 resolve docker.io/library/python:3.11-slim@sha256:6d98ca198cea726f2c86da2699594339a7b7ff08e49728797b4ed6e3b5c3b62a 0.0s done
#5 DONE 0.0s

#6 [2/5] WORKDIR /app
#6 CACHED

#7 [4/5] RUN pip install --no-cache-dir -r requirements.txt
#7 CACHED

#8 [3/5] COPY requirements.txt .
```

Jenkins / FoodExpress-Pipeline #8

```
#8 [3/5] COPY requirements.txt .
#8 CACHED

#9 [5/5] COPY . .
#9 CACHED

#10 exporting to image
#10 exporting layers done
#10 exporting manifest sha256:c402c447069ebb198b5d37fc19b942f2c48a122f793eae74058a50c67587ad03 done
#10 exporting config sha256:dcae8e48d7bb9da67fae157fdbd36f8dd2f986f953b3c9ddc56aa9e9387b9217 done
#10 exporting attestation manifest sha256:a6f9b49505bed4fce76d21563463fc2b7e754e326aff269bb1623ce9e01eb777 0.0s done
#10 exporting manifest list sha256:017c57cab3e0b43af716fa4539782b42c744cd0d2c43b7b66748da1edflee76a
#10 exporting manifest list sha256:017c57cab3e0b43af716fa4539782b42c744cd0d2c43b7b66748da1edflee76a 0.0s done
#10 naming to docker.io/foodexpress/fastapi:v1.0 done
#10 unpacking to docker.io/foodexpress/fastapi:v1.0 0.0s done
#10 DONE 0.1s
[Pipeline] echo (hide)
Build Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Image As Container)
[Pipeline] echo
Deploying container...
[Pipeline] sh
+ docker stop foodexpress-container
foodexpress-container
+ docker rm foodexpress-container
foodexpress-container
+ docker run --name foodexpress-container -d -p 8000:8000 foodexpress/fastapi:v1.0
cdd4f19f3960192ec73a435bc6488644142eaea20aa30e29bd61316aa91509be
[Pipeline] echo
FoodExpress FastAPI is running inside Docker on port 8000
[Pipeline] }
[Pipeline] // stage
```

Jenkins / FoodExpress-Pipeline #8

```
Build Done
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Image As Container)
[Pipeline] echo
Deploying container...
[Pipeline] sh
+ docker stop foodexpress-container
foodexpress-container
+ docker rm foodexpress-container
foodexpress-container
+ docker run --name foodexpress-container -d -p 8000:8000 foodexpress/fastapi:v1.0
cdd4f19f3960192ec73a435bc6488644142eaea20aa30e29bd61316aa91509be
[Pipeline] echo
FoodExpress FastAPI is running inside Docker on port 8000
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful! FoodExpress APIs are live.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.2

Jenkins / FoodExpress-Pipeline #8 / Stages

#8

Rerun

Started by Hello

Started 6 min 44 sec ago

Queued 3 ms

Took 3.6 sec

Graph

Start

Checkout SCM

Clone

Copy

Build

Run Image As...

Post Actions

End

Search

Run Image As Container

0.83s

Started 6m 50s ago

Jenkins

Checkout SCM

0.22s

Clone

0.25s

Copy

0.33s

Build

0.84s

Run Image As Container

0.83s

Post Actions

71ms

Deploying container...

15ms

docker stop foodexpress-container || true docker rm foodexpress-co...

0.77s

FoodExpress FastAPI is running inside Docker on port 8000

14ms

FoodExpress FastAPI is running inside Docker on port 8000

Jenkins 2.541.2

```
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-20-96:~$ sudo docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
cdd4f19f3960   foodexpress/fastapi:v1.0   "uvicorn main:app --..."   8 minutes ago   Up 8 minutes   0.0.0.0:8000->8000/tcp, :::8000->8000/tcp   foodexpress-container
ubuntu@ip-172-31-20-96:~$
```

Task 9 - After Manual Build - Access your GET API using POSTMAN and EC2 Public IP. - Screenshot

My Workspace

Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

GET http://54.91.214.38:8000/orders

GET http://54.91.214.38:8000/orders

Send

Body

```
1 {
2   "orders": [
3     {
4       "id": 1,
5       "customer": "Alice",
6       "item": "Burger",
7       "quantity": 2,
8       "status": "pending"
9     },
10    {
11      "id": 2,
12      "customer": "Bob",
13      "item": "Pizza",
14      "quantity": 1,
15      "status": "delivered"
16    },
17    {
18      "id": 3,
19      "customer": "Charlie",
20      "item": "Sushi",
21      "quantity": 3,
22      "status": "preparing"
23    }
24  ],
25  "total": 3
26 }
```

200 OK • 563 ms • 378 B

Start using Agent Mode!
Your plan includes 50 AI credits per month to use Agent Mode.

GET http://54.91.214.38:8000/health

Describe what you need. Press @ for context, / for Skills.

Auto

My Workspace

Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

GET http://54.91.214.38:8000/health

GET http://54.91.214.38:8000/health

Send

Body

```
1 {
2   "status": "healthy"
3 }
```

200 OK • 562 ms • 145 B

Start using Agent Mode!
Your plan includes 50 AI credits per month to use Agent Mode.

My Workspace

Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

GET http://54.91.214.38:8000/search

GET http://54.91.214.38:8000/search

Send

Body

```
1 {
2   "Inside search endpoint": "This is a placeholder for search functionality"
3 }
```

200 OK • 560 ms • 200 B

Start using Agent Mode!
Your plan includes 50 AI credits per month to use Agent Mode.

My Workspace

Work with your local codebase
Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

GET http://54.91.214.38:8000/orders/1

GET http://54.91.214.38:8000/orders/1

Send

Body

```
1 {
2   "id": 1,
3   "customer": "Alice",
4   "item": "Burger",
5   "quantity": 2,
6   "status": "pending"
7 }
```

200 OK • 560 ms • 200 B

Start using Agent Mode!
Your plan includes 50 AI credits per month to use Agent Mode.

Task 10 - Configure Webhooks in GitHub and check its linked or not - Screenshot

The screenshot shows the GitHub repository settings for 'devops-assignment-3' under the 'Webhooks' tab. The left sidebar lists various settings categories: General, Access, Collaborators, Moderation options, Code and automation, Branches, Tags, Rules, Actions, Models, Webhooks (selected), Copilot, Environments, Codespaces, Pages, Security, Advanced Security, Deploy keys, Secrets and variables, Integrations, and GitHub Apps. The main content area is titled 'Webhooks / Add webhook' and contains the following fields and options:

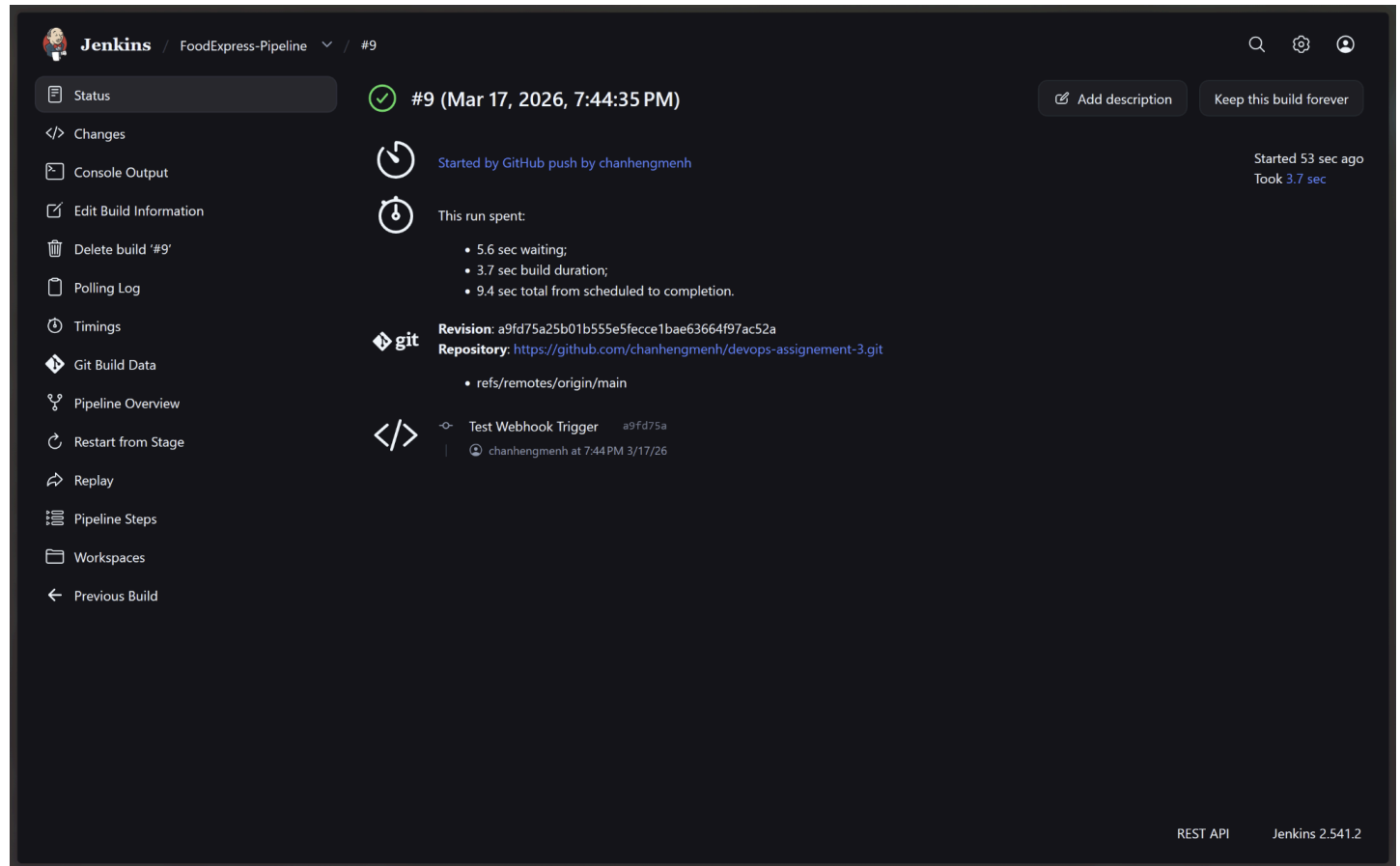
- Payload URL ***: `http://54.91.214.38:8080/github-webhook/`
- Content type ***: `application/x-www-form-urlencoded`
- Secret**: (Empty text field)
- SSL verification**: ☒ Enable SSL verification ☐ Disable (not recommended)
- Which events would you like to trigger this webhook?**: ☒ Send me everything. ☐ Just the push event. ☐ Let me select individual events.
- Active**: ☒ Active. Below it, a note says: 'We will deliver event details when this hook is triggered.'

At the bottom of the form is a blue button labeled 'Add webhook'.

→ make a new commit

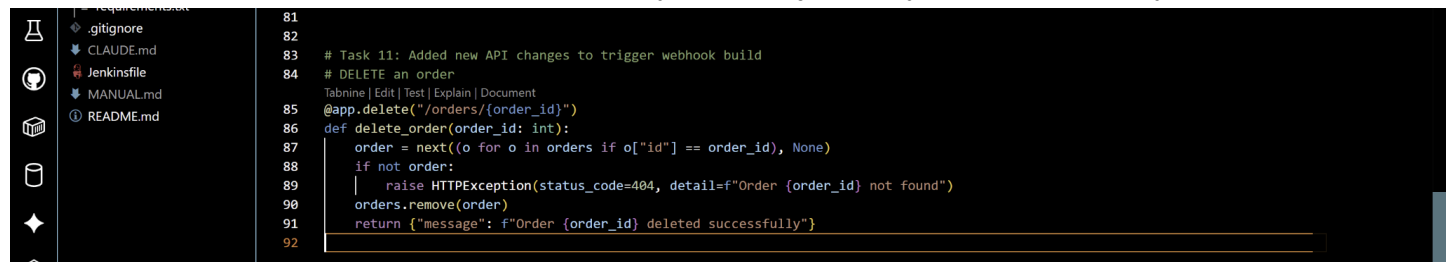
```
00ab9751..0e2d37 main -> main
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git add .\README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git commit -m 'Test Webhook Trigger'
[main a9fd75a] Test Webhook Trigger
 1 file changed, 31 insertions(+), 25 deletions(-)
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 638 bytes | 638.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com-personal:chanhengmenh/devops-assignment-3.git
 de82d37..a9fd75a main -> main
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3>
```


→ Triggered:



The screenshot shows the Jenkins web interface for a build named '#9' in the 'FoodExpress-Pipeline' project. The build status is 'Success' (green checkmark) and it was triggered by a 'GitHub push by chanhengmenh' on March 17, 2026, at 7:44:35 PM. The build took 3.7 seconds. The left sidebar contains navigation links: Status, Changes, Console Output, Edit Build Information, Delete build '#9', Polling Log, Timings, Git Build Data, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main area shows the build details, including the revision 'a9fd75a25b01b555e5fecce1bae63664f97ac52a' and the repository 'https://github.com/chanhengmenh/devops-assignment-3.git'. The build steps are listed, with 'Test Webhook Trigger' being the current step, initiated by 'chanhengmenh' at 7:44 PM on 3/17/26. The bottom right corner indicates 'REST API' and 'Jenkins 2.541.2'.

Task 11 - Add new Api like post or put or delete in your index.js file in your local repository. - Screenshot

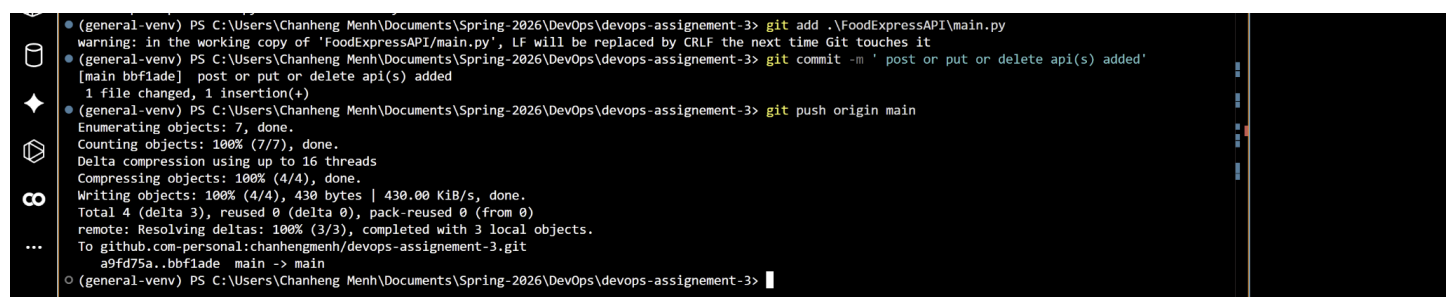


The screenshot shows a code editor with a file named 'index.js'. The code defines a REST API endpoint for deleting an order. The file content is as follows:

```
81
82
83 # Task 11: Added new API changes to trigger webhook build
84 # DELETE an order
85 @app.delete("/orders/{order_id}")
86 def delete_order(order_id: int):
87     order = next((o for o in orders if o["id"] == order_id), None)
88     if not order:
89         raise HTTPException(status_code=404, detail=f"Order {order_id} not found")
90     orders.remove(order)
91     return {"message": f"Order {order_id} deleted successfully"}
92
```

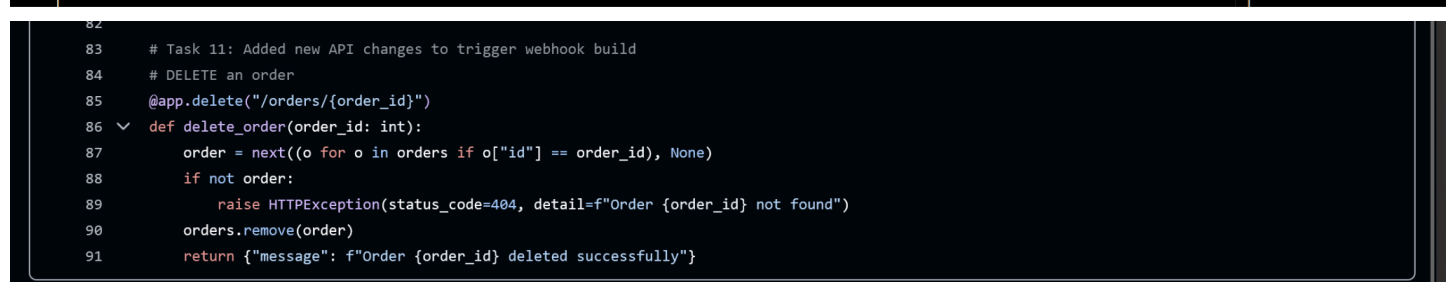
Task 12 - Add and commit it.

Task 13 - PUSH it in the GitHub. - Screenshot



The screenshot shows a terminal window with the following commands and output:

```
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git add .\FoodExpressAPI\main.py
warning: in the working copy of 'FoodExpressAPI/main.py', LF will be replaced by CRLF the next time Git touches it
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git commit -m 'post or put or delete api(s) added'
[main bbflade] post or put or delete api(s) added
1 file changed, 1 insertion(+)
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 430 bytes | 430.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com-personal:chanhengmenh/devops-assignment-3.git
a9fd75a..bbflade main -> main
(general-venv) PS C:\Users\Chanheng Menh\Documents\Spring-2026\DevOps\devops-assignment-3>
```



The screenshot shows a code editor with a file named 'index.js'. The code defines a REST API endpoint for deleting an order. The file content is as follows:

```
82
83 # Task 11: Added new API changes to trigger webhook build
84 # DELETE an order
85 @app.delete("/orders/{order_id}")
86 def delete_order(order_id: int):
87     order = next((o for o in orders if o["id"] == order_id), None)
88     if not order:
89         raise HTTPException(status_code=404, detail=f"Order {order_id} not found")
90     orders.remove(order)
91     return {"message": f"Order {order_id} deleted successfully"}

```

Task 14 - Access your new API (post or put or delete) using POSTMAN and EC2 Public IP. - Screenshot

→ Existing Data:

HTTP **http://54.91.214.38:8000/orders** Save Share

GET **http://54.91.214.38:8000/orders** Send

Docs Params Authorization Headers 8 **Body** Scripts Settings Cookies

Body Cookies Headers 4 Test Results **200 OK** • 544 ms • 453 B • 🌐 • ⋮

{ } JSON Preview Visualize

```
1 {
2   "orders": [
3     {
4       "id": 1,
5       "customer": "Alice",
6       "item": "Burger",
7       "quantity": 2,
8       "status": "pending"
9     },
10    {
11      "id": 2,
12      "customer": "Bob",
13      "item": "Pizza",
14      "quantity": 1,
15      "status": "delivered"
16    },
17    {
18      "id": 3,
19      "customer": "Charlie",
20      "item": "Sushi",
21      "quantity": 3,
22      "status": "preparing"
23    },
24    {
25      "id": 5,
26      "customer": "Diana",
27      "item": "Tacos",
28      "quantity": 2,
29      "status": "pending"
30    }
31  ],
32  "total": 4
33 }
```

→ New API: DELETE

Q Search Invite Upgrade ⏏ ⚙️ 🔍

DEL **http://54.91.214.38:8000/orders/5** GET **http://54.91.214.38:8000/orders/5** No enviro... Save Share

DELETE **http://54.91.214.38:8000/orders/5** Send

Docs Params Authorization **Headers 8** Body Scripts Settings Cookies

Body Cookies Headers 4 Test Results **200 OK** • 538 ms • 167 B • 🌐 • ⋮

{ } JSON Preview Visualize

```
1 {
2   "message": "Order 5 deleted successfully"
3 }
```

New Chat + ⌚

Start using Agent Mode!
Your plan includes 50 AI credits per month to use Agent Mode.

→ Test Existing API:

My Workspace

Work with your local codebase

Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

Search

DEL http://54.91.214.38:80C POST http://54.91.214.38:80C

http://54.91.214.38:8000/orders

POST http://54.91.214.38:8000/orders

Docs Params Authorization Headers 8 Body Scripts Settings

Body

```
1 {
2   "customer": "Chanheng",
3   "item": "Gum",
4   "quantity": 2,
5   "status": "pending"
6 }
```

Body Cookies Headers 4 Test Results

201 Created 539 ms 255 B

```
1 {
2   "message": "Order created successfully",
3   "order": {
4     "id": 6,
5     "customer": "Chanheng",
6     "item": "Gum",
7     "quantity": 2,
8     "status": "pending"
9   }
10 }
```

New Chat

Start using Agent Mode!

Your plan includes 50 AI credits per month to use Agent Mode.

My Workspace

Work with your local codebase

Postman updates collections, specs and environments, and follows git branches. Push to cloud when ready. [Learn more](#)

Open folder

Search

DEL http://54.91.214.38:80C PUT http://54.91.214.38:80C

http://54.91.214.38:8000/orders/6

PUT http://54.91.214.38:8000/orders/6

Docs Params Authorization Headers 8 Body Scripts Settings

Body

```
1 {
2   "customer": "Chanheng Menh",
3   "item": "Gum & PC Components",
4   "quantity": 6,
5   "status": "Done!"
6 }
```

Body Cookies Headers 4 Test Results

200 OK 547 ms 269 B

```
1 {
2   "message": "Order updated successfully",
3   "order": {
4     "id": 6,
5     "customer": "Chanheng Menh",
6     "item": "Gum & PC Components",
7     "quantity": 6,
8     "status": "Done!"
9   }
10 }
```

New Chat

Start using Agent Mode!

Your plan includes 50 AI credits per month to use Agent Mode.

Full Pipeline script:



```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Clone') {
7       steps {
8         echo 'Cloning the code from GitHub...'
9         git url: 'https://github.com/chanhengmenh/devops-assignment-3.git', branch: 'main'
10        echo 'Cloning Done'
11      }
12    }
13
14    stage('Copy') {
15      steps {
16        echo 'Copying files to /home/ubuntu/current...'
17        sh 'cp -r FoodExpressAPI /home/ubuntu/current'
18        echo 'Copy Done'
19      }
20    }
21
22    stage('Build') {
23      steps {
24        echo 'Building Docker image...'
25        sh 'docker build -t foodexpress/fastapi:v1.0 /home/ubuntu/current/FoodExpressAPI'
26        echo 'Build Done'
27      }
28    }
29
30    stage('Run Image As Container') {
31      steps {
32        echo 'Deploying container...'
33        sh '''
34          docker stop foodexpress-container || true
35          docker rm foodexpress-container || true
36          docker run --name foodexpress-container -d -p 8000:8000 foodexpress/fastapi:v1.0
37          ...
38        '''
39        echo 'FoodExpress FastAPI is running inside Docker on port 8000'
40      }
41    }
42  }
43
44  post {
45    failure {
46      echo 'Pipeline failed. Check the Console Output for details.'
47    }
48    success {
49      echo 'Deployment Successful! FoodExpress APIs are live.'
50    }
51  }
52 }
53
```